

Align Technology

NASDAQ : \$12.46B mkt cap · 71M sh · \$1.1B cash, zero debt · \$1.10B cash, no debt (net cash, p_fail ~0); ~71.3M shares; ~\$200M buyback active;
ALGN · Revenue \$4.035B FY25 (+1%, flat-lining); GAAP op margin compressed to 13.5% (non-GAAP ~21%); FCF ~\$0.5B; ~70% gross margin;
Mature- ~\$175, YTD ~+7-8%, 52-wk \$122-\$208 - deeply de-rated from the 2021 ~\$700 peak (fwd P/E ~15)
Company DCF

\$174.84

THE CENTRAL QUESTION

Does clear-aligner category growth and Invisalign share retention outrun ASP/share erosion from Angelalign and at-home commoditization?

Align trades at ~\$12.5B (~\$174.84, deeply de-rated from the 2021 ~\$700 peak to ~\$175) - ~25x FCF on \$4.04B revenue growing low-single-digit, ~70% gross margin, net cash and a buyback. It is the category-defining clear-aligner brand (Invisalign + iTero) in a market still only ~15-20% penetrated, but growth has flat-lined (+1%) as discretionary demand softens and Angelalign/at-home rivals pressure ASPs (-1 to -2% guided). The DCF asks whether category penetration plus the teen/Palatal-Expander TAM outrun price/share erosion. The finding lands ~fair (-1.5%): the modal case is +1%, de-rated but not cheap.

PROBABILITY-WEIGHTED EXPECTED VALUE

\$172.27 expected fair value today
-1.5% vs spot \$174.84

FORWARD COMPOUNDED VALUE

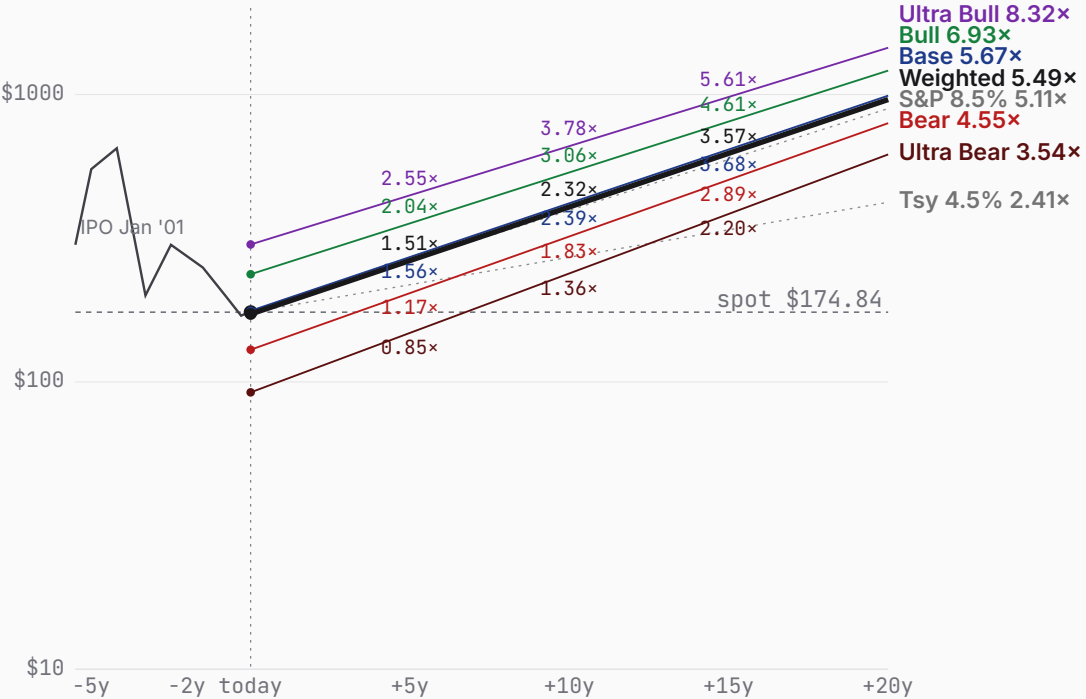
+5y	+10y	+15y	+20y
\$264.49	\$406.29	\$624.48	\$960.41
1.51x spot	2.32x spot	3.57x spot	5.49x spot

WEIGHTING RATIONALE

Ultra Bear 14% Demand stalls; ASP/share erode; ~\$92 (-47%).
Bear 27% Low-single-digit grind; ASP pressure caps it; ~\$129 (-26%).
Base 34% Category penetration resumes; margin firms; ~\$177 (+1%).
Bull 17% Teen/IPE TAM expands; share holds; ~\$237 (+35%).
Ultra Bull 8% Category compounds; Invisalign re-rates; ~\$301 (+72%).

Bull (17%) + Ultra Bull (8%) together contribute \$64.33 — 37% of the \$172.27 expected value despite 25% combined probability. Today's spot (\$174.84) sits 1% above the weighted expected; forward-weighted value crosses spot between +5y and +10y.

PRICE + PROJECTIONS · HISTORICAL THROUGH PROBABILITY-WEIGHTED FORWARD



ULTRA BEAR	\$91.97	BEAR	\$129.43	BASE	\$176.81	BULL	\$236.91	ULTRA BULL	\$300.75
	-47.4% vs spot		-26.0% vs spot		+1.1% vs spot		+35.5% vs spot		+72.0% vs spot
CAGR +1.4% WACC 10.0% CAGR +1.4% Prob 14%		CAGR +3.8% WACC 9.5% CAGR +3.8% Prob 27%		CAGR +6.6% WACC 9.0% CAGR +6.6% Prob 34%		CAGR +8.4% WACC 8.5% CAGR +8.4% Prob 17%		CAGR +10.0% WACC 8.2% CAGR +10.0% Prob 8%	
Demand stalls; ASP/share erode.		Low-single-digit grind; ASP pressure caps it.		Penetration resumes; margin firms.		Teen/IPE TAM expands; share holds.		Category compounds; Invisalign re-rates.	



Align Technology scenario narratives

PROBABILITY WEIGHTS

Ultra Bear 14% Bear 27% Base 34% Bull 17% Ultra Bull 8% · Weighted expected \$172.27 (-1.5% vs spot)

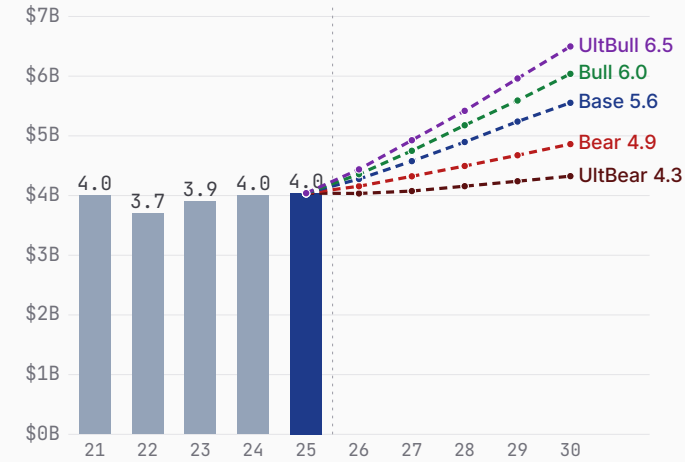
ULTRA BEAR	\$91.97	BEAR	\$129.43	BASE	\$176.81	BULL	\$236.91	ULTRA BULL	\$300.75
	-47.4% vs spot		-26.0% vs spot		+1.1% vs spot		+35.5% vs spot		+72.0% vs spot
Probability 14%		Probability 27%		Probability 34%		Probability 17%		Probability 8%	
Demand stalls; ASP/share erode.		Low-single-digit grind; ASP pressure caps it.		Penetration resumes; margin firms.		Teen/IPE TAM expands; share holds.		Category compounds; Invisalign re-rates.	
WHY 14%		WHY 27%		WHY 34%		WHY 17%		WHY 8%	
The genuine bear: revenue is already flat (+1% FY25), op margin has slid four straight years (24.7%→13.5%), ASPs are guided down, and Angelalign is suing on patents while taking China share. 14% weight on price/share erosion beating category growth.		Share retention continues but the discretionary backdrop + ASP pressure + Chinese competition cap the multiple. 27% as the plausible 'holds the lead but doesn't convert to growth' path, consistent with the flat top line and guided ASP declines.		Requires penetration-led case growth + the teen/IPE TAM to offset ASP erosion - consistent with the record Q1 case volume (686,000/+6.7%) and reaffirmed guidance, but not a step-change re-acceleration. 34% as the central, near-fair outcome.		The IPE clearance + record teen case volume + a still-underpenetrated category make a genuine re-acceleration credible. ~17% weight on the TAM expansion landing while share holds.		Everything works - penetration AND the teen/IPE TAM AND share defense AND a margin/multiple re-rate, the high-quality-compounder outcome. ~8%, the long-duration tail.	
WHAT HAPPENS		WHAT HAPPENS		WHAT HAPPENS		WHAT HAPPENS		WHAT HAPPENS	
The discretionary big-ticket consumer stays soft and Angelalign/at-home commoditization wins on price: case volume flat-lines, ASPs keep falling beyond the guided -1 to -2%, and a richer low-cost mix de-levers a ~70% gross margin into low-double-digit operating margins. Revenue is essentially flat; the category-penetration story stalls while share leaks.		Category penetration keeps volume rising but ASP/mix erosion and a competitive market hold growth to ~4% and margins below the historical peak. Invisalign defends its lead without re-accelerating; teen/IPE helps at the edges but doesn't change the trajectory. Revenue compounds ~4% at a low-teens FCF margin.		The modal path: clear-aligner penetration of the ~15-20%-penetrated ortho market resumes mid-single-digit case growth, the teen/Palatal-Expander TAM adds incremental volume, and iTero Lumina anchors the installed base - blending to ~7% revenue growth at a firming ~16-18% operating margin as ASP pressure is offset by volume. Revenue compounds ~7% toward ~\$5.5B.		Clear-aligner category penetration accelerates and the teen/Palatal-Expander launch opens the larger teen ortho TAM (the majority of global case-starts), while Invisalign holds share through patents + the iTero installed base. Volume growth and operating leverage push margins back toward 20%. Revenue compounds ~8-9%.		The full second act: clear aligners take a decisive share of the global ortho market, teen/IPE becomes a durable growth leg, iTero scan-data + AI deepen the cornered resource, and Align re-rates to a high-quality medtech compounder at ~70% gross margin. Revenue compounds ~10% to peak margins.	
A category leader whose volume and price are both pressured earns a low-growth ~13x terminal multiple → DCF ~\$92 (-47%). Net cash sets a floor but can't offset a stalled top line and a margin that keeps grinding lower.		DCF ~\$129 (-26%) - a defended-but-decelerating franchise at ~15x FCF is worth below today's price; the market's discount is warranted if growth stays mid-single-digit and ASPs keep slipping.		DCF ~\$177 (+1%) - essentially at spot. A category-defining brand with a long penetration runway is genuinely valuable, but at ~16x FCF and mid-single-digit growth the modal case only recovers to about today's price; the de-rate corrected the 2021 excess to fair, not to a bargain.		DCF ~\$237 (+35%) - if the penetration + teen-TAM engine re-accelerates and ASPs stabilize, the de-rated entry re-rates with the fundamentals; this is the core of the bull.		DCF ~\$301 (+72%) - the brand reclaims a premium medtech multiple as category growth + share defense + margin recovery all land; ~8% probability, the asymmetric upside the depressed entry is buying.	

Align Technology business snapshot

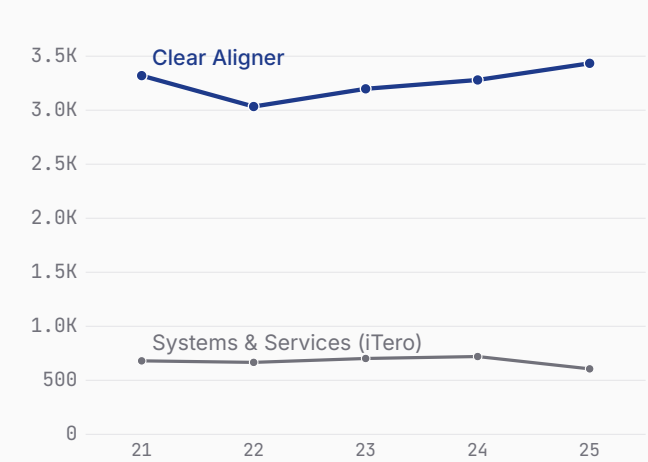
Bear \$129.43 Base \$176.81 Bull \$236.91

FY21-FY25 history + FY26-FY30 scenario projections · fiscal years end late Dec · Q1 FY2026 (Apr 2026) results

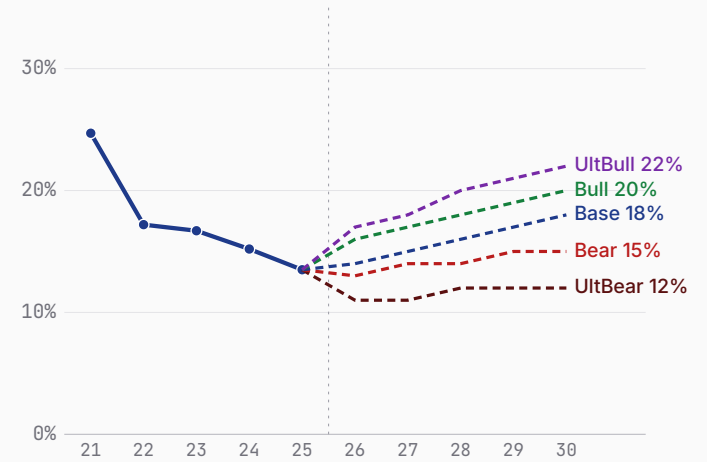
Revenue (\$B) — history + scenarios to FY30



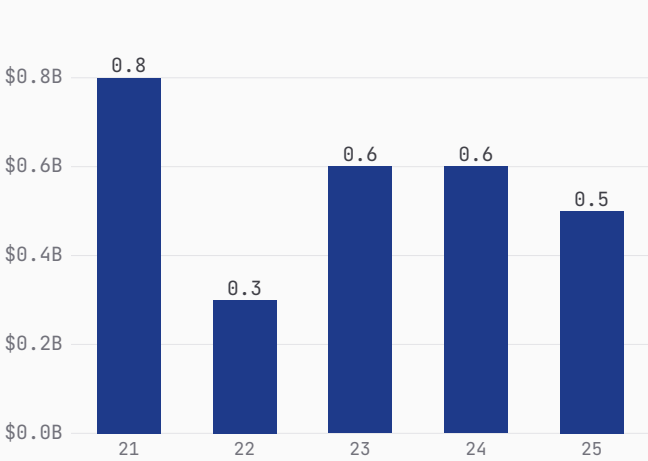
Revenue by segment (\$M)



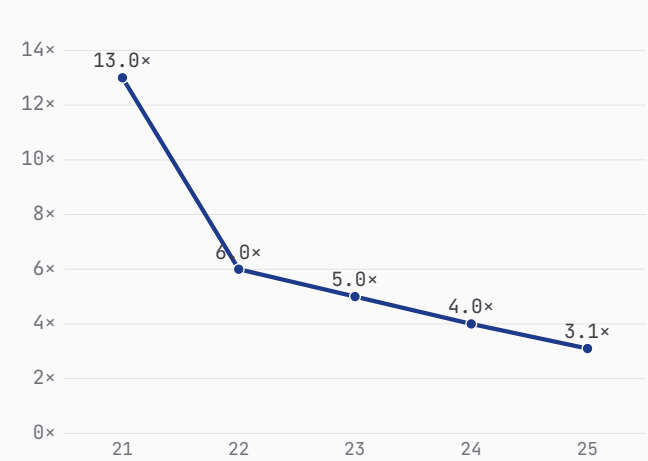
Op margin — history + scenarios



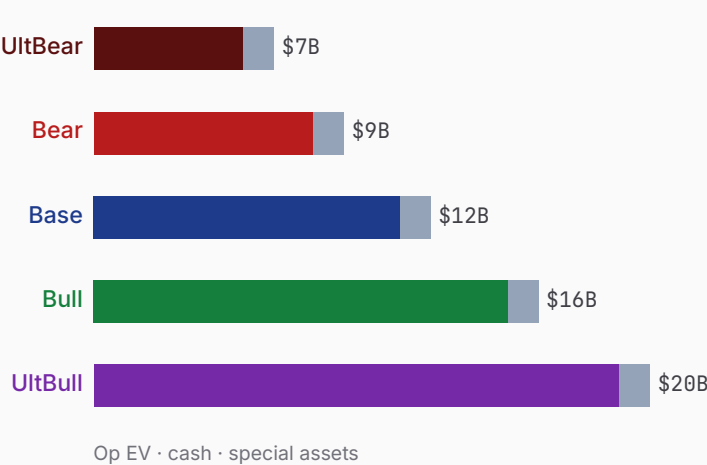
Free cash flow (\$B) — history



EV / Revenue — historical



Equity build (Op EV + net cash)



Sources: SEC XBRL company facts (CIK 1097149), Align Q1 FY2026 release (revenue \$1,040M/+6.2%, Clear Aligner \$856M, iTero/Systems \$184M, record 686,000 Invisalign cases/+6.7%, gross margin 70.8%, FY26 ASPs guided -1 to -2%). Revenue is a growth path off FY25; FCF = revenue x FCF margin; Gordon terminal. Multiples vs Dentsply, Straumann, Angelalign.

Align Technology 7 Powers · the moat, scored

Bear \$129.43 Base \$176.81 Bull \$236.91

Arena. Clear-aligner orthodontics / dental devices - Align Technology (Invisalign, the category-defining brand, + iTero scanners) vs Angelalign, SmileDirect-style at-home/DIY, Dentsply Sirona, Straumann/ClearCorrect, and traditional wires/brackets; auditing a brand + cornered-resource Power against commoditization and a discretionary-demand market. **POWER AUDIT** dominant-power durability: medium

POWER AUDIT · 7 POWERS

Scale Economies ■■■	Largest clear-aligner manufacturer; manufacturing/R&D/sales leverage at \$4B revenue and ~70% gross margin - real, but Angelalign is scaling fast in China.
Network Economies ■■■	Invisalign-trained doctor base + iTero/exocad workflow create two-sided pull (more doctors → more patients), but not a hard network effect.
Counter-Positioning ■■■	Was the disruptor of wires/brackets; now the incumbent being counter-positioned by lower-cost Angelalign and at-home/DIY models.
Switching Costs ■■■	Doctors invested in iTero hardware + Invisalign training face friction switching systems; patient-side switching is low (a single discretionary purchase).
Branding · thesis leans here ■■■	The dominant Power: Invisalign is the category-defining consumer brand with pricing power and direct-to-consumer demand pull - but eroding at the ASP line under competition.
Cornered Resource ■■■	iTero installed base (>125,000), accumulated scan-data, and a deep patent estate (the basis of the Angelalign suits/ITC action) - durable, the hardest piece to replicate.
Process Power ■■■	Mass-customization manufacturing + the digital treatment-planning stack; a real learning curve, modest as a standalone Power.

COMPETITIVE READ

Align's brand Power is genuine - Invisalign is the category-defining clear-aligner name with pricing power and consumer demand pull, reinforced by the iTero installed base and patent estate as a cornered resource - but the franchise is decelerating: revenue is flat (+1%), op margin has slid four straight years to 13.5%, and ASPs are guided down under Angelalign + at-home price competition. The Audit: branding is durable (medium) but eroding at the price line; the cornered resource (iTero/scan-data/patents) is the real defense, and the live threats are commoditization + a discretionary-demand market. The ~75% de-rate from the 2021 peak corrected the prior excess to roughly fair, not to a bargain; the DCF tests whether category penetration + the teen/IPE TAM outrun ASP/share erosion, and the modal case lands +1% - de-rated, but priced about right.

PEER SET

Angelalign Fast-growing Chinese clear-aligner challenger; competes on price, taking China share - the live patent/ITC threat.	30% gr · 25% mgn · HK-listed
Dentsply Sirona Diversified dental incumbent (incl. SureSmile aligners + imaging); the legacy-dental comp.	0% gr · 10% mgn · ~12x EV/EBITDA
Straumann / ClearCorrect Premium dental/implant group with a growing aligner line; the high-end aligner challenger.	10% gr · 25% mgn · ~30x P/E
At-home / DIY (SmileDirect legacy) Direct-to-consumer aligners at the low end; SmileDirect went bankrupt, but the price-competitive segment persists.	· private

THREAT VECTORS · FALSIFIERS

Branding · Angelalign + at-home/DIY (price competition) falsifier: ASPs fall faster than the guided -1 to -2% as the brand premium erodes against lower-cost aligners.
Cornered Resource · Angelalign (patents / ITC) falsifier: The patent suits + ITC import-block action fail, letting competitors replicate the technology and scan-data edge.
Scale Economies · Discretionary-demand softness + China bulk-procurement falsifier: Case volume flat-lines or declines and a low-cost mix de-levers the ~70% gross margin below ~13% operating margin.

Align Technology show your work

Bear \$129.43 Base \$176.81 Bull \$236.91 · Weighted \$172.27 (-1.5%)

ULTRA BEAR					\$91.97					BEAR					\$129.43					BASE					\$176.81					BULL					\$236.91					ULTRA BULL					\$300.75				
ASSUMPTIONS										ASSUMPTIONS										ASSUMPTIONS										ASSUMPTIONS										ASSUMPTIONS									
5y revenue CAGR (%)					+1.4					5y revenue CAGR (%)					+3.8					5y revenue CAGR (%)					+6.6					5y revenue CAGR (%)					+8.4					5y revenue CAGR (%)					+10.0				
Year-5 op margin (%)					12.0					Year-5 op margin (%)					15.0					Year-5 op margin (%)					18.0					Year-5 op margin (%)					20.0					Year-5 op margin (%)					22.0				
WACC (%)					10.0					WACC (%)					9.5					WACC (%)					9.0					WACC (%)					8.5					WACC (%)					8.2				
Terminal growth (%)					2.0					Terminal growth (%)					2.5					Terminal growth (%)					2.5					Terminal growth (%)					3.0					Terminal growth (%)					3.0				
5y revenue CAGR (%)					+1.4					5y revenue CAGR (%)					+3.8					5y revenue CAGR (%)					+6.6					5y revenue CAGR (%)					+8.4					5y revenue CAGR (%)					+10.0				
Starting revenue (\$B)					4.04					Starting revenue (\$B)					4.04					Starting revenue (\$B)					4.04					Starting revenue (\$B)					4.04					Starting revenue (\$B)					4.04				
Scenario probability (%)					14					Scenario probability (%)					27					Scenario probability (%)					34					Scenario probability (%)					17					Scenario probability (%)					8				
5-YEAR DCF · \$B										5-YEAR DCF · \$B										5-YEAR DCF · \$B										5-YEAR DCF · \$B										5-YEAR DCF · \$B									
FY	Rev	Margin	FCF	PV FCF	FY	Rev	Margin	FCF	PV FCF	FY	Rev	Margin	FCF	PV FCF	FY	Rev	Margin	FCF	PV FCF	FY	Rev	Margin	FCF	PV FCF	FY	Rev	Margin	FCF	PV FCF	FY	Rev	Margin	FCF	PV FCF															
FY26	4.04	+11%	+0.40	+0.37	FY26	4.16	+13%	+0.46	+0.42	FY26	4.28	+14%	+0.51	+0.47	FY26	4.36	+16%	+0.57	+0.52	FY26	4.44	+17%	+0.62	+0.57	FY26	4.44	+17%	+0.62	+0.57	FY26	4.44	+17%	+0.62	+0.57															
FY27	4.08	+11%	+0.41	+0.34	FY27	4.32	+14%	+0.52	+0.43	FY27	4.58	+15%	+0.59	+0.50	FY27	4.75	+17%	+0.67	+0.56	FY27	4.93	+18%	+0.74	+0.63	FY27	4.93	+18%	+0.74	+0.63	FY27	4.93	+18%	+0.74	+0.63															
FY28	4.16	+12%	+0.46	+0.34	FY28	4.50	+14%	+0.54	+0.41	FY28	4.90	+16%	+0.69	+0.53	FY28	5.18	+18%	+0.78	+0.61	FY28	5.42	+20%	+0.87	+0.68	FY28	5.42	+20%	+0.87	+0.68	FY28	5.42	+20%	+0.87	+0.68															
FY29	4.24	+12%	+0.47	+0.32	FY29	4.67	+15%	+0.61	+0.42	FY29	5.24	+17%	+0.73	+0.52	FY29	5.59	+19%	+0.90	+0.65	FY29	5.96	+21%	+1.01	+0.74	FY29	5.96	+21%	+1.01	+0.74	FY29	5.96	+21%	+1.01	+0.74															
FY30	4.32	+12%	+0.48	+0.30	FY30	4.86	+15%	+0.63	+0.40	FY30	5.55	+18%	+0.83	+0.54	FY30	6.04	+20%	+0.97	+0.64	FY30	6.50	+22%	+1.17	+0.79	FY30	6.50	+22%	+1.17	+0.79	FY30	6.50	+22%	+1.17	+0.79															
Σ PV explicit FCF					+1.66					Σ PV explicit FCF					+2.08					Σ PV explicit FCF					+2.56					Σ PV explicit FCF					+2.98					Σ PV explicit FCF					+3.42				
+ PV terminal (g 2.0%)					3.77					+ PV terminal (g 2.5%)					5.88					+ PV terminal (g 2.5%)					8.54					+ PV terminal (g 3.0%)					12.03					+ PV terminal (g 3.0%)					15.63				
EQUITY BUILD · \$B & PER SHARE										EQUITY BUILD · \$B & PER SHARE										EQUITY BUILD · \$B & PER SHARE										EQUITY BUILD · \$B & PER SHARE										EQUITY BUILD · \$B & PER SHARE									
Operating EV					\$5.43B					Operating EV					\$7.96B					Operating EV					\$11.10B					Operating EV					\$15.01B					Operating EV					\$19.05B				
+ Cash & investments					\$1.10B					+ Cash & investments					\$1.10B					+ Cash & investments					\$1.10B					+ Cash & investments					\$1.10B					+ Cash & investments					\$1.10B				
= Equity value					\$6.53B					= Equity value					\$9.06B					= Equity value					\$12.20B					= Equity value					\$16.11B					= Equity value					\$20.15B				
FY30 shares (M)					71					FY30 shares (M)					70					FY30 shares (M)					69					FY30 shares (M)					68					FY30 shares (M)					67				
= Expected per share					\$91.97					= Expected per share					\$129.43					= Expected per share					\$176.81					= Expected per share					\$236.91					= Expected per share					\$300.75				
FUTURE FAIR VALUE · IF SCENARIO PLAYS OUT										FUTURE FAIR VALUE · IF SCENARIO PLAYS OUT										FUTURE FAIR VALUE · IF SCENARIO PLAYS OUT										FUTURE FAIR VALUE · IF SCENARIO PLAYS OUT										FUTURE FAIR VALUE · IF SCENARIO PLAYS OUT									
+5y	+10y	+15y	+20y		+5y	+10y	+15y	+20y		+5y	+10y	+15y	+20y		+5y	+10y	+15y	+20y		+5y	+10y	+15y	+20y		+5y	+10y	+15y	+20y		+5y	+10y	+15y	+20y																
\$148.12	\$238.55	\$384.18	\$618.73		\$203.75	\$320.76	\$504.95	\$794.91		\$272.04	\$418.57	\$644.03	\$990.92		\$356.23	\$535.65	\$805.43	\$1,211.09		\$446.01	\$661.42	\$980.88	\$1,454.62		\$446.01	\$661.42	\$980.88	\$1,454.62		\$446.01	\$661.42	\$980.88	\$1,454.62																
0.85×	1.36×	2.20×	3.54×		1.17×	1.83×	2.89×	4.55×		1.56×	2.39×	3.68×	5.67×		2.04×	3.06×	4.61×	6.93×		2.55×	3.78×	5.61×	8.32×		2.55×	3.78×	5.61×	8.32×		2.55×	3.78×	5.61×	8.32×																

Align Technology

supporting analysis · disclaimers · glossary

Bear \$129.43 Base \$176.81 Bull \$236.91

PUSHBACK · WHY THE BASE CASE IS TOO HARSH

- 1

The de-rate corrected an excess, it didn't create a bargain.
Down ~75% from the 2021 peak looks cheap, but at ~25x FCF and low-single-digit growth the modal DCF is +1% - growth, not the multiple, is the constraint.
- 2

The category is still only ~15-20% penetrated.
Clear aligners are early in displacing wires/brackets; teen/IPE opens the larger teen ortho TAM the flat top line masks.
- 3

Brand + installed base are the moat.
Invisalign is category-defining; iTero (>125,000 scanners) + scan-data + the patent estate are hardest for Angelalign to replicate.
- 4

ASP/share erosion is the whole question.
ASPs guided -1 to -2%, Angelalign taking China share + litigating patents, at-home at the low end - 'category grows' vs 'Align grows' is the fork.
- 5

Net cash, p_fail ~0.
\$1.10B cash, no debt, ~\$200M buyback - no solvency risk; the downside is growth/multiple, not survival.

FALSIFICATION TRIGGERS

Bull validation

Invisalign case volume re-accelerates above ~+7% · teen/IPE case-starts scale materially · ASPs stabilize (not down) · operating margin recovers back toward ~20%

Bear validation

case volume flat-lines or declines for 2+ quarters · ASPs fall faster than the guided -1 to -2% · Angelalign/at-home take visible share · op margin grinds below ~13%

Reframe needed

if clear aligners commoditize (Angelalign wins on price + the ITC/patent actions fail), the branding + cornered-resource Powers erode - re-rate the terminal toward a low-growth, price-taking dental-device multiple

DISCLAIMERS · PLEASE READ BEFORE USING THIS DOCUMENT

Not investment advice.

This document is produced for entertainment and educational purposes only by an individual amateur investor. "Alameda Research 2: Electric Boogaloo" is the unincorporated personal concept of one person and is not a registered investment management entity. Nothing herein constitutes investment advice, a recommendation, or a solicitation to buy or sell any security.

Not a professional.

The author is not a registered investment advisor, broker-dealer, CFA charterholder, certified financial planner, or licensed financial professional of any kind. The author has no formal training in equity research or capital markets. No fiduciary relationship is created by reading this document.

AI-assisted analysis.

Substantial portions of this analysis were produced with the assistance of large language models. LLMs are known to fabricate facts, miscalculate numbers, hallucinate sources, and present incorrect information with high apparent confidence. Every figure, claim, and projection in this document should be independently verified before acting on it.

Forward-looking statements.

Scenarios, valuations, projections, and any forward-looking statements involve substantial assumptions and uncertainty. Actual results may differ materially from those projected. Past performance is not indicative of future results. The model is a model; reality is not.

Author may hold positions.

The author may own, intend to acquire, or hold short exposure to \$ALGN or related securities at any time. Positions may change without notice and without an update to this document. Do not assume the author's positions match the tone of the analysis.

Do your own research.

Do not make investment decisions on the basis of this document. Consult a licensed financial professional and read the company's official SEC filings (10-K, 10-Q, 8-K, proxy) before forming any investment view. If you wouldn't trust your retirement to a chatbot, don't trust this either.

GLOSSARY · CONCEPTS REFERENCED IN THE NARRATIVE

Clear Aligner — Align's largest segment (~82-85% of revenue) - Invisalign removable aligners that displace traditional wires/brackets; case volume x ASP is the swing factor.

iTero / Systems & Services — Intraoral scanners (iTero Lumina, installed base >125,000) + software/services - the cornered-resource base that feeds Invisalign workflow and scan-data.

ASP — Average selling price per case; guided -1 to -2% in FY26 on country/product mix and competition - the price half of the volume-vs-price question.

IPE (Invisalign Palatal Expander) — US-FDA-cleared device targeting teen/kids case-starts (the majority of global ortho) - the TAM-expansion lever in the bull.

FCF margin — Free cash flow / revenue (~12-13% FY25); the mature-DCF lever - whether category growth converts the ~70% gross margin to free cash flow as ASP pressure bites.